Geocomputation

Spatial Autocorrelation



Module outline

W1	Reproducible Spatial Analysis	Core Spatial Analysis
W2	Spatial Queries and Geometric Operations	
W3	Point Pattern Analysis	
W4	Spatial Autocorrelation	
W5	Spatial Models	
W6	Raster Data Analysis	Applied Spatial Analysis
W7	Geodemographic Classification	
W8	Accessibility Analysis	
W9	Beyond the Choropleth	Data Visualisation
W10	Complex Visualisations	

This week

- Spatial dependence
- Measuring spatial autocorrelation
- Spatial weights matrix

Spatial dependence

Tobler's first law of geography

"Everything is related to everything else, but near things are more related than distant things."

Waldo Tobler, 1970

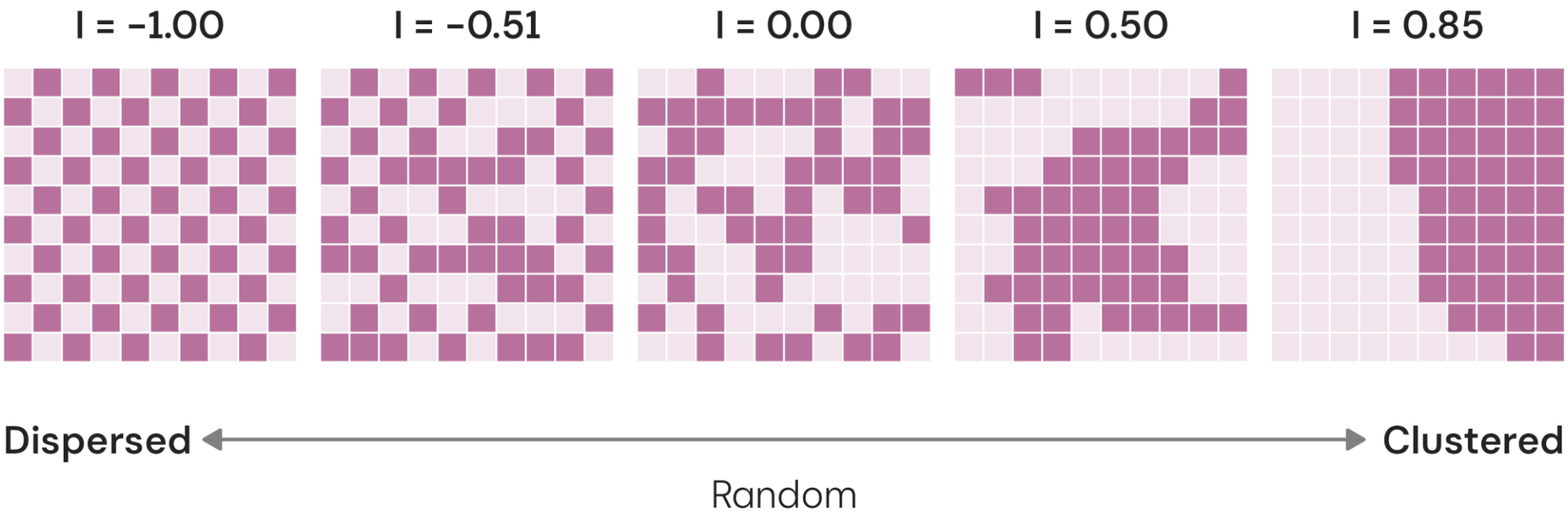
Spatial dependence

- Spatial dependence refers to the concept that the value of a variable at one location is influenced, to some extent, by the value of the same variable at nearby locations.
- This is often understood through the concept of distance decay, where the influence decreases as distance increases.

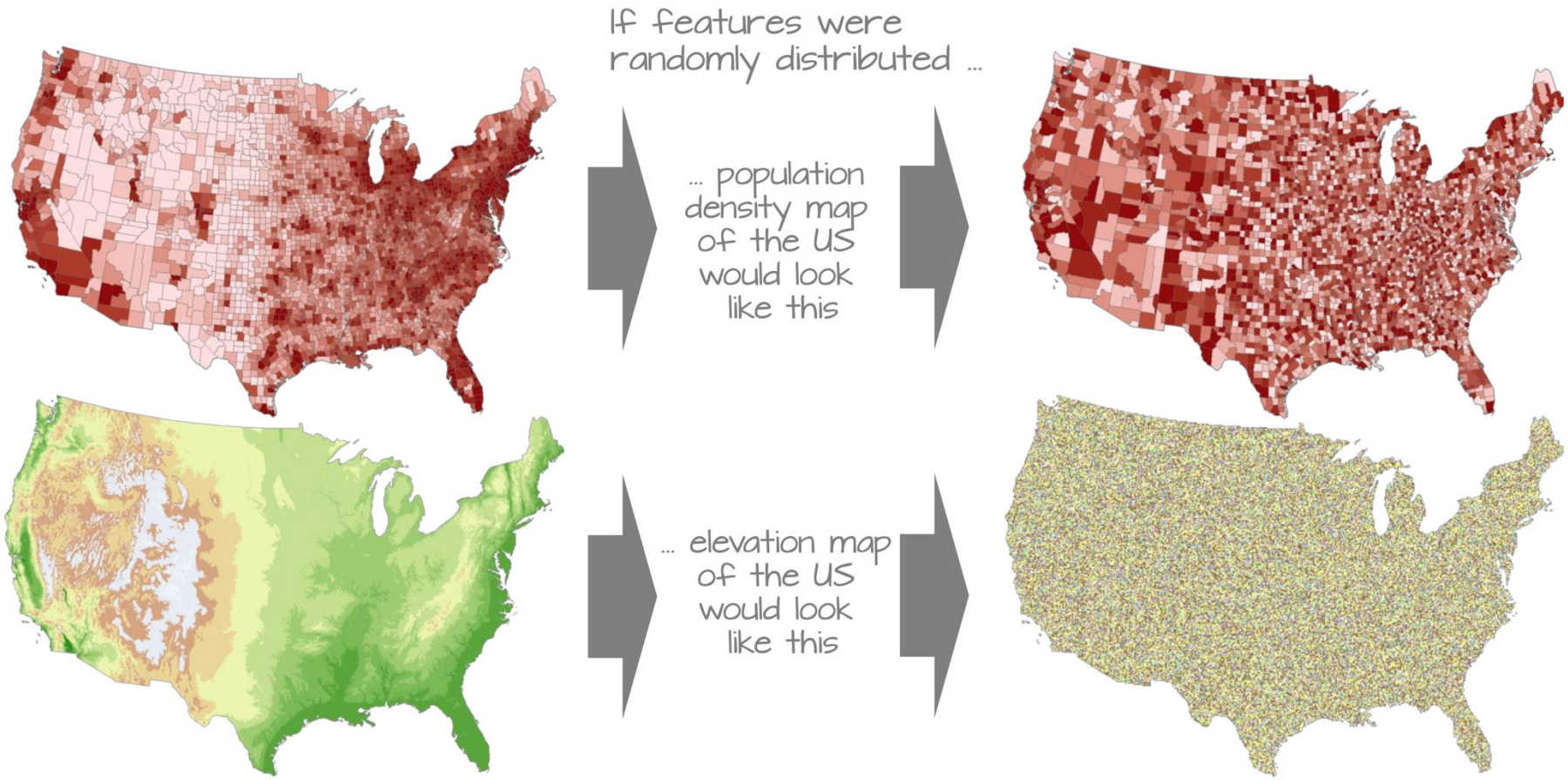
What is spatial autocorrelation?

- Spatial autocorrelation formalises spatial dependence: it measures the degree to which similar values cluster together in space.
- Local measures show where high values and low values are concentrated (hotspots and coldspots).
- No spatial autocorrelation means complete spatial randomness (CSR): where a value ends up tells us nothing about the values around it.

From dispersed to clustered



Real patterns are not random



Gimond, M. 2025. Intro to GIS and Spatial Analysis. [online]
https://mgimond.github.io/Spatial/index.html

Global and local measures

Spatial Autocorrelation can be measured in two ways:

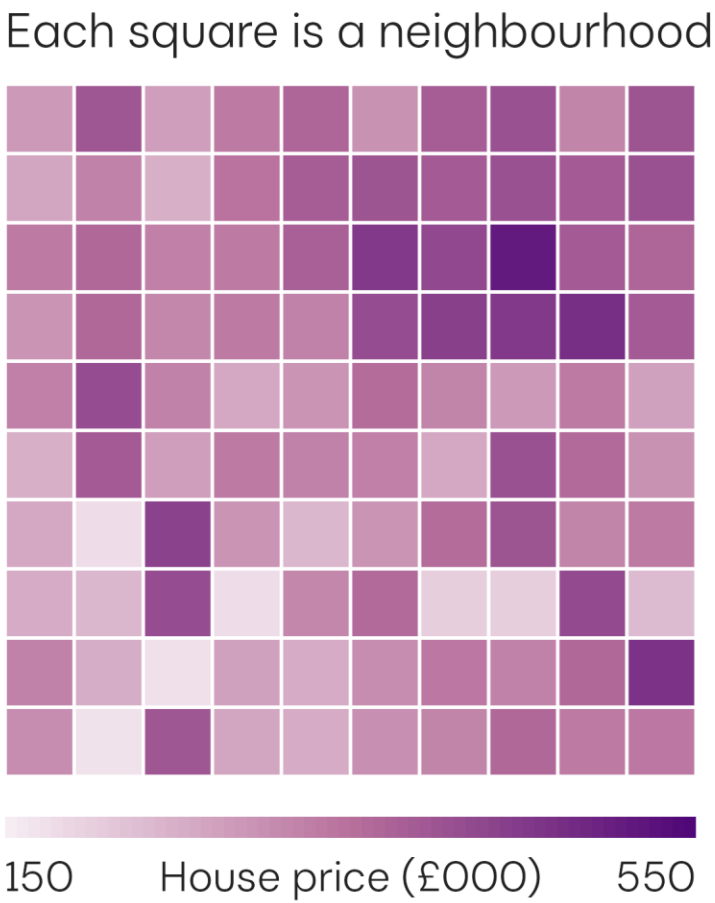
1) Global Spatial Autocorrelation: This assesses the overall spatial dependence across the entire dataset.
2) Local Spatial Autocorrelation: This focuses on the differences between each unit of analysis and its neighbours.

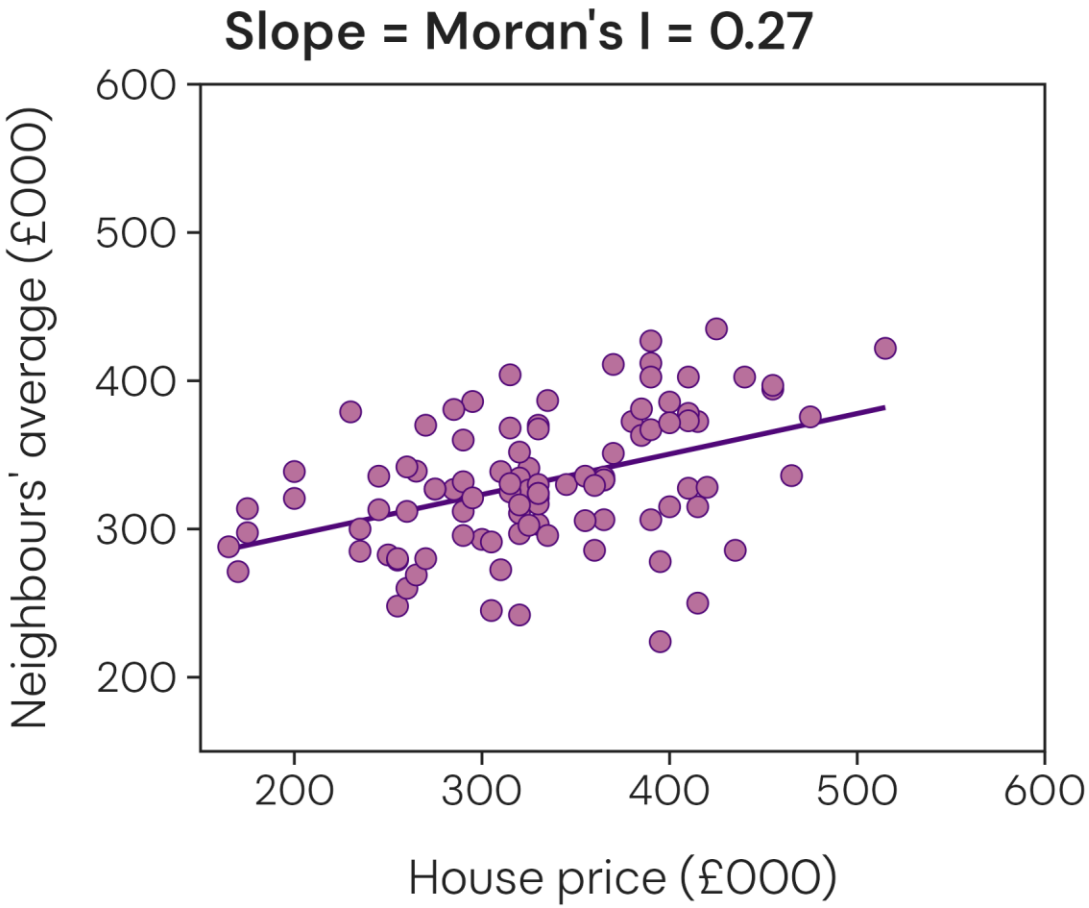
Global spatial autocorrelation

Moran's I

- Moran's I: The most commonly used indicator of global spatial autocorrelation.
- Identifies neighbours for each target feature (e.g. polygon) and summarises their values by computing their means to create a spatially lagged variable value.
- Plots the target feature's value against its spatially lagged mean value and fits a linear model to the points.
- The slope of the fitted line (β estimate) is the Moran's I statistic.

The Moran scatterplot





Simulated example: house prices in 100 neighbourhoods.

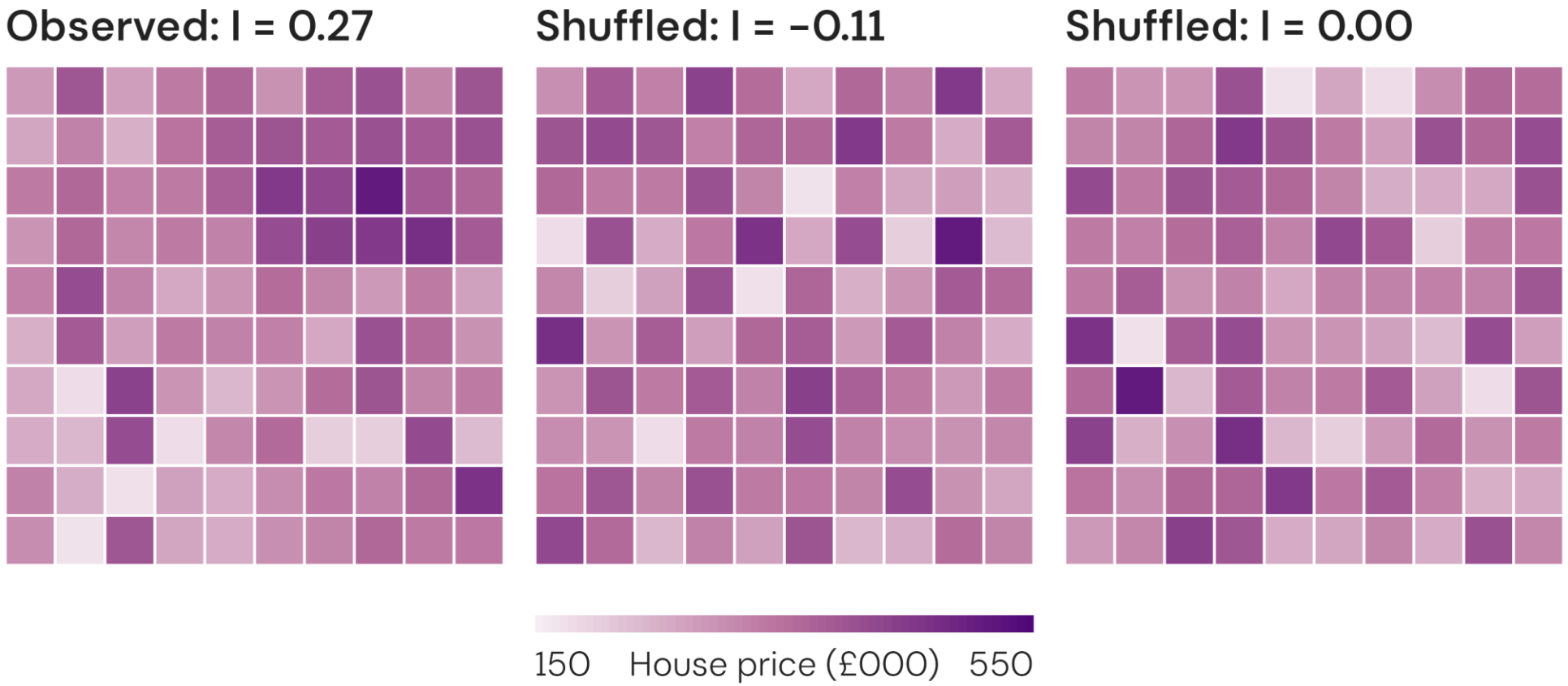
Moran's I

- The Moran's I statistic typically ranges from −1 to +1:
  - **+1** Indicates perfect clustering (positive spatial autocorrelation).
  - **0** Suggests a random pattern (no spatial autocorrelation).
  - **−1** Indicates perfect dispersion (negative spatial autocorrelation).
- How to assess the significance of the relationship?

Moran's I

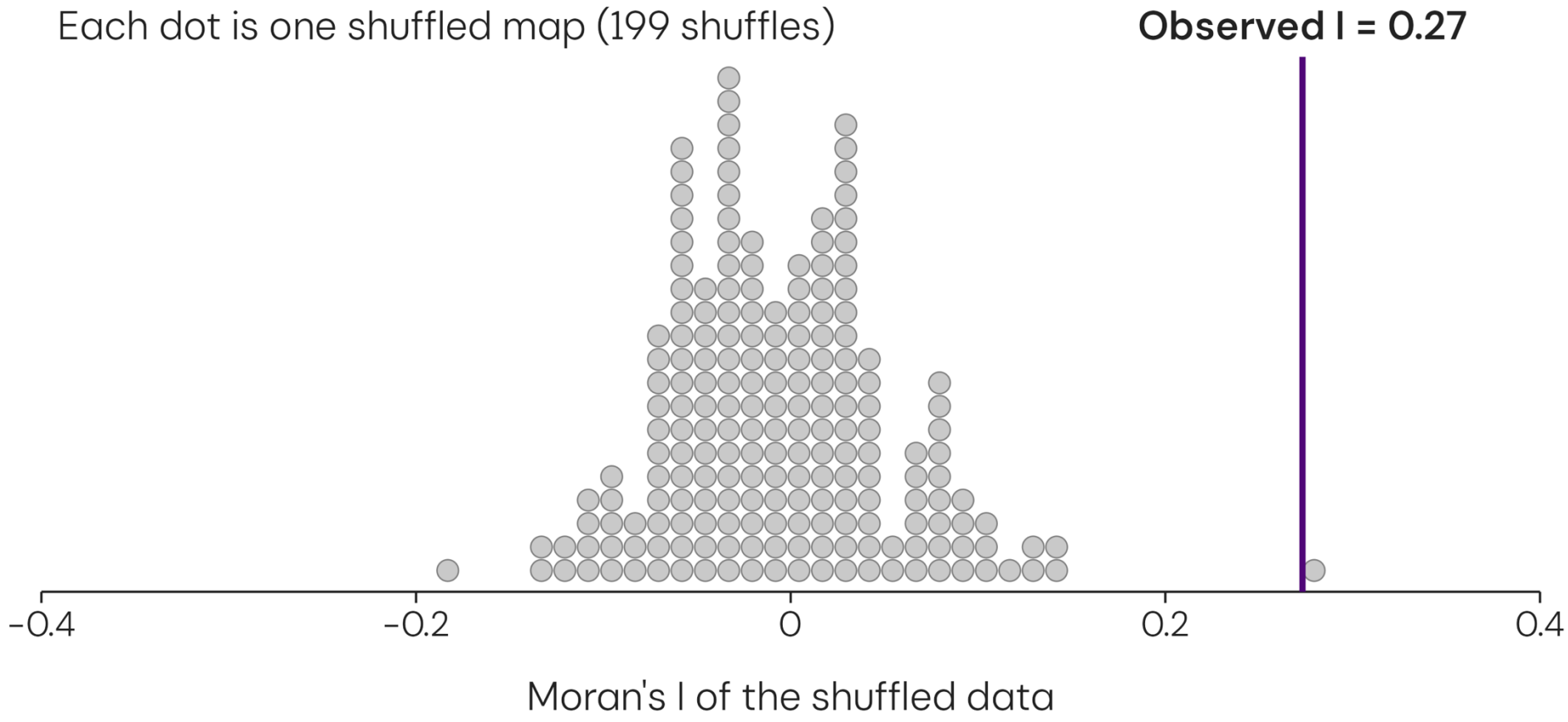
- To understand whether our relationship is significant, we can use either an analytical approach or a computational approach. The latter is the preferred option as it does not require making any assumption about the shape and layout of our data set – for this we can use a Monte Carlo test.
- This approach randomly and repeatedly assigns values to polygons in the data set.
- The output is a sampling distribution of Moran's I values under the (null) hypothesis that attribute values are randomly distributed across the study area.

Shuffle the values, recalculate I



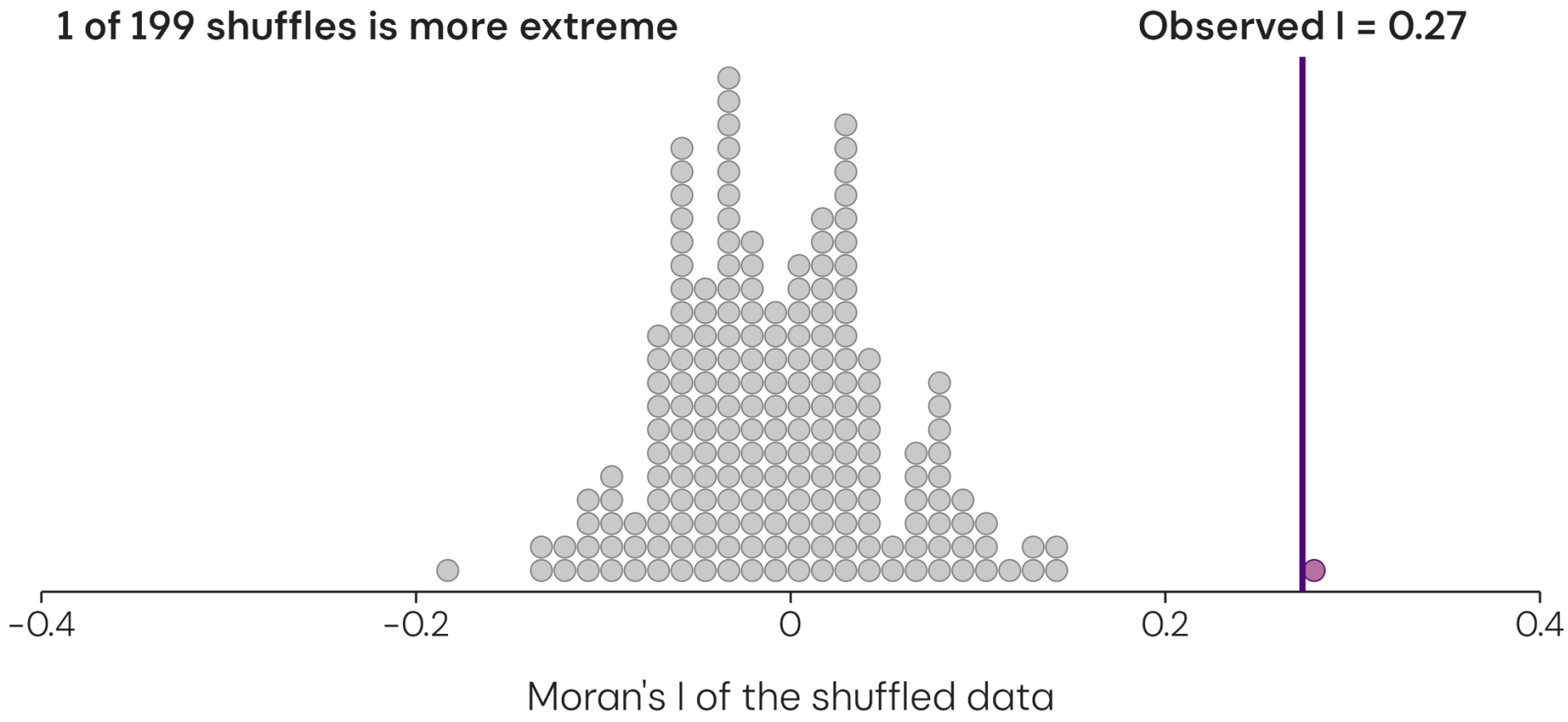
Simulated example: house prices in 100 neighbourhoods.

Moran’s I of 199 shuffled maps



Simulated example: house prices in 100 neighbourhoods, 199 shuffles.

How extreme is the observed value?



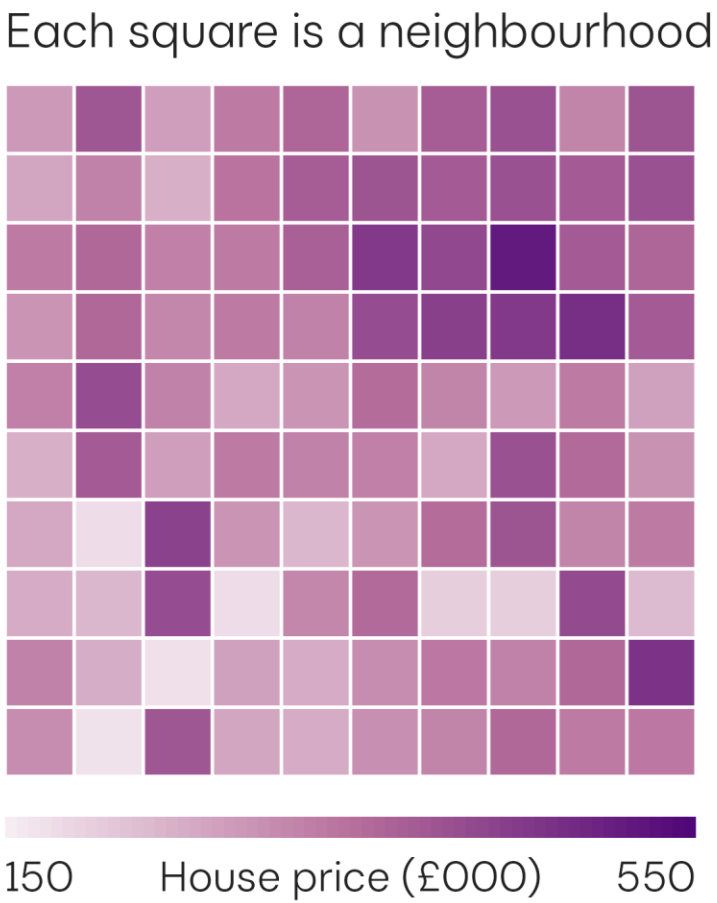
Simulated example: house prices in 100 neighbourhoods, 199 shuffles.

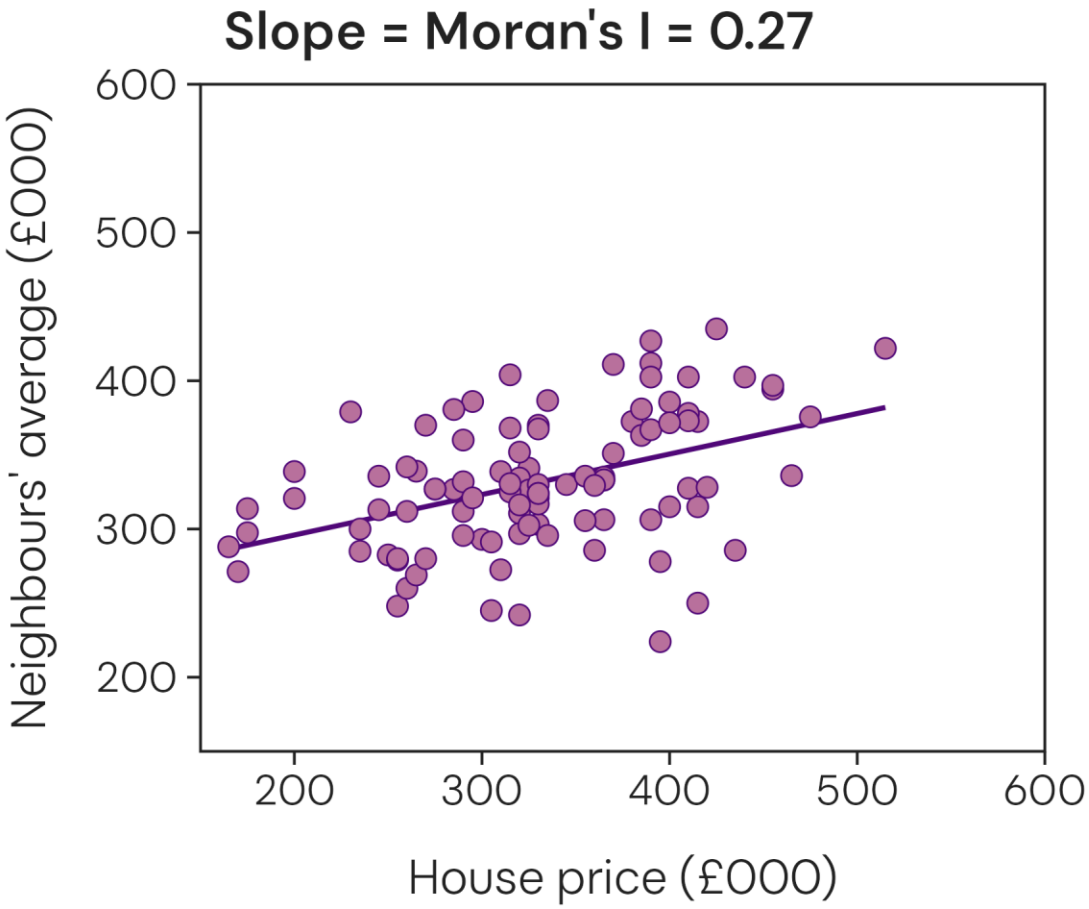
Moran's I

- A pseudo *p*-value is generated from the simulation results.
- For instance: if out of 199 simulations, just one simulation result is more extreme than our observed statistic, *p* is equal to (1 + 1) / (199 + 1) = 0.01. This means that, if the values were randomly distributed across space, we would see a Moran's I at least this extreme in only 1% of cases.
- Be aware that the pseudo *p*-value is only a summary of the results from the reference distribution and should not be interpreted as an analytical *p*-value (which assumes normality).

Local spatial autocorrelation

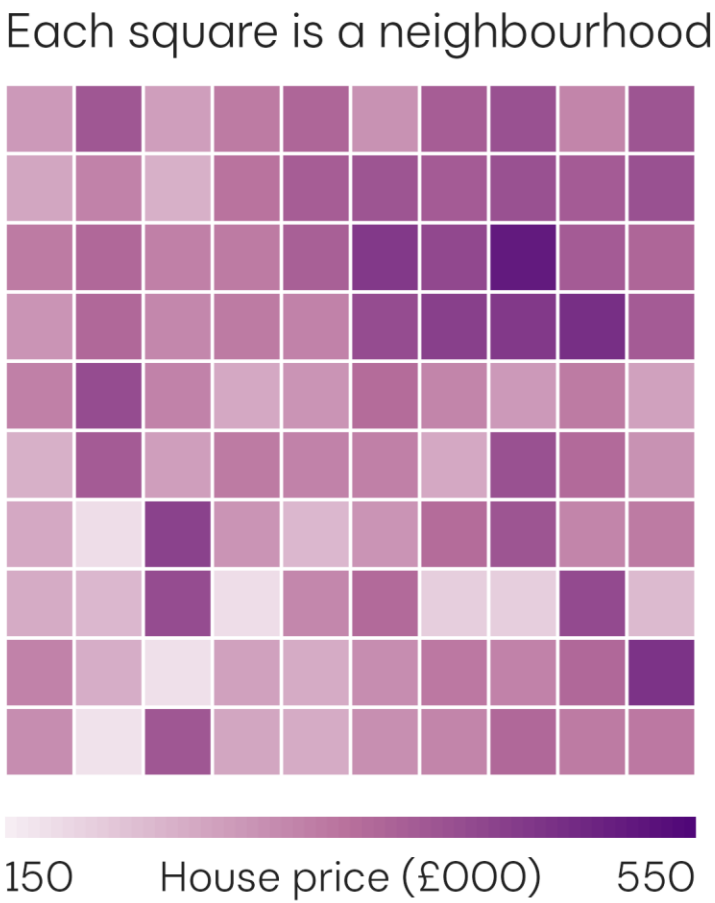
Local Moran's I

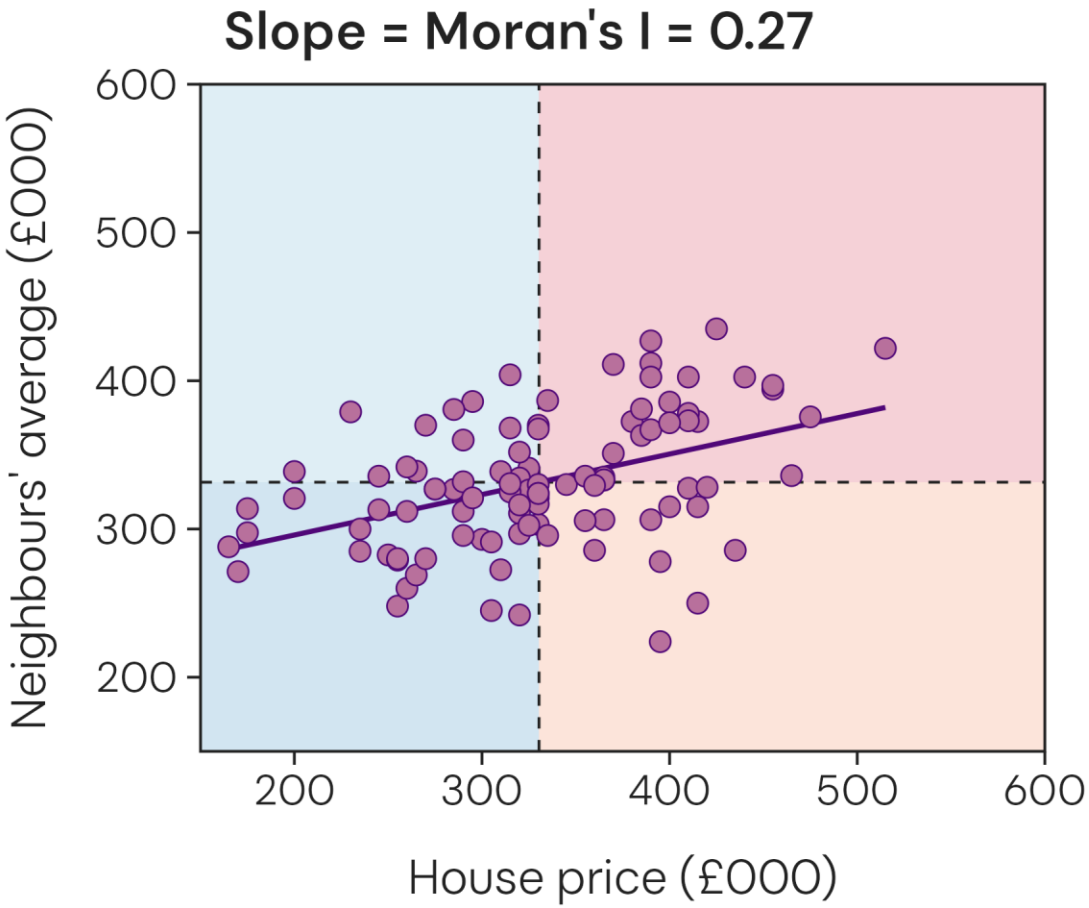




Simulated example: house prices in 100 neighbourhoods.

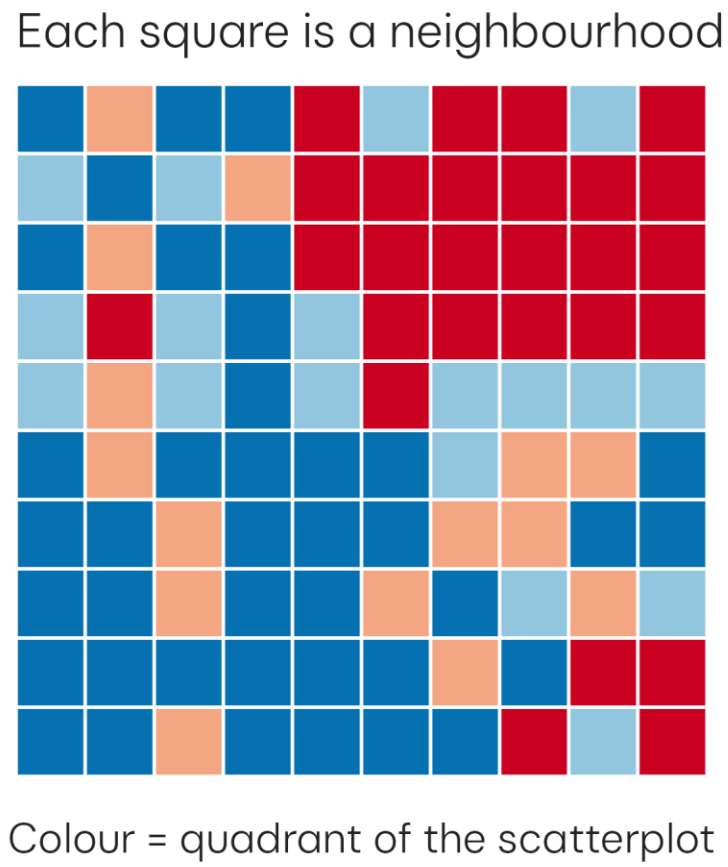
Local Moran's I: the four quadrants

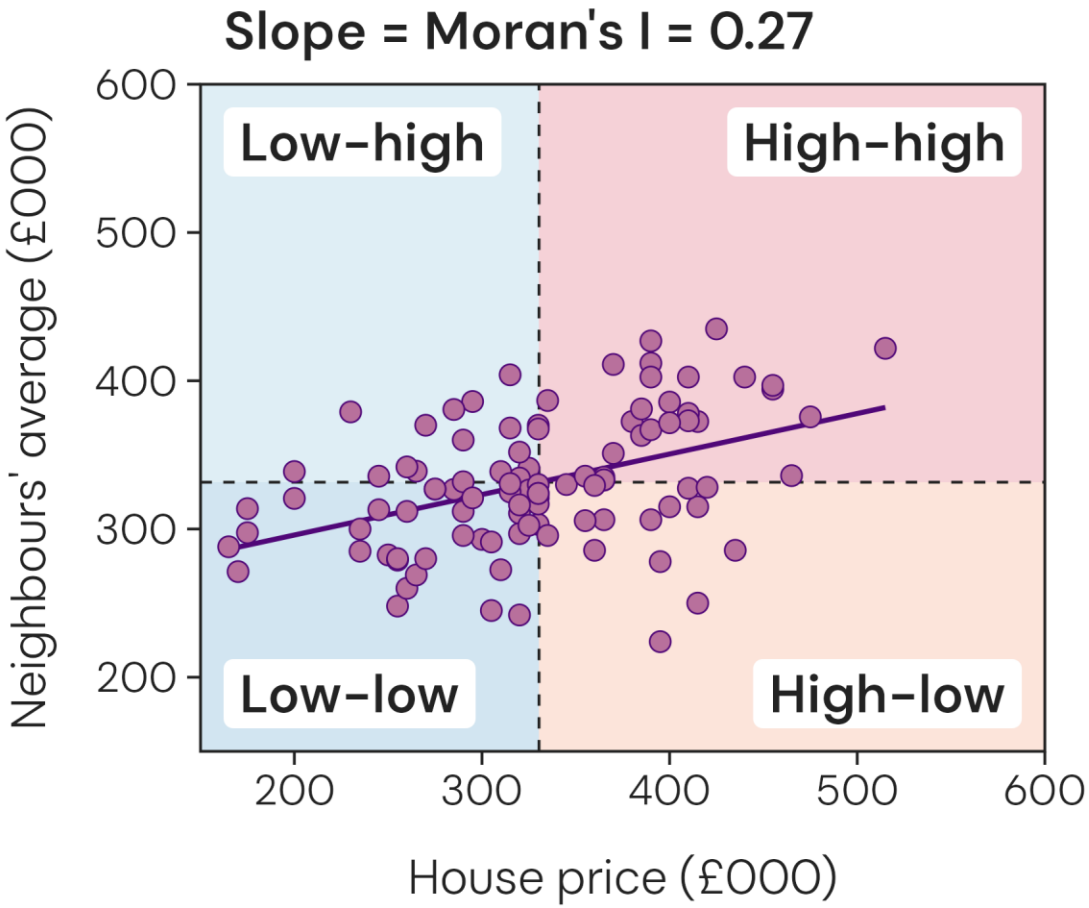




Simulated example: house prices in 100 neighbourhoods.

Local Moran's I: the four quadrants



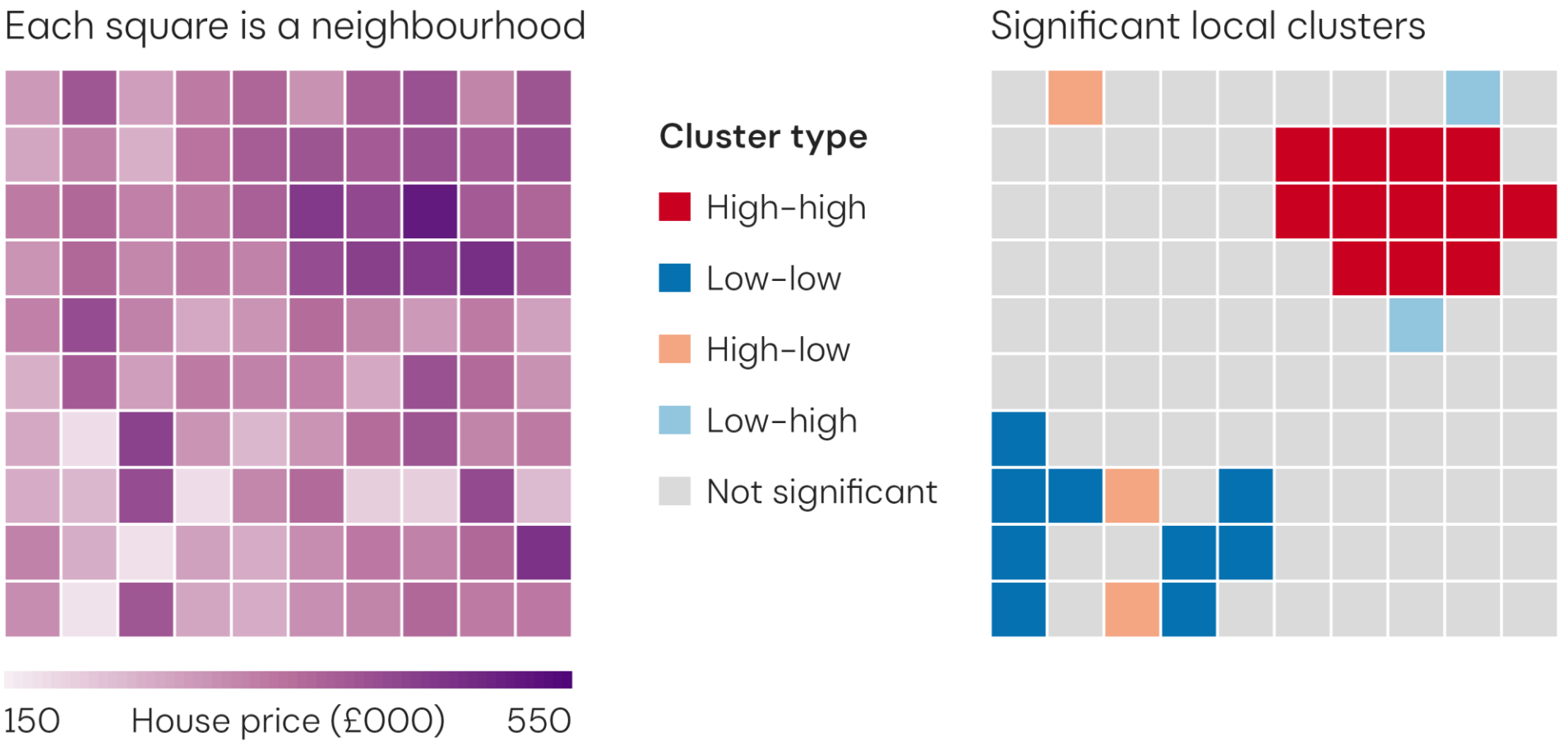


Simulated example: house prices in 100 neighbourhoods.

Local Moran's I

- Decomposing the Moran's I statistic: Local Moran's I
- Assesses spatial autocorrelation at the local level by evaluating each feature and its surrounding neighbourhood.
- Four cluster types: high-high and low-low clusters, and high-low and low-high outliers.
- Commonly known as cluster and outlier analysis.
- Monte Carlo simulation assesses the significance of every area, so with many areas some will look significant by chance.

Mapping the clusters



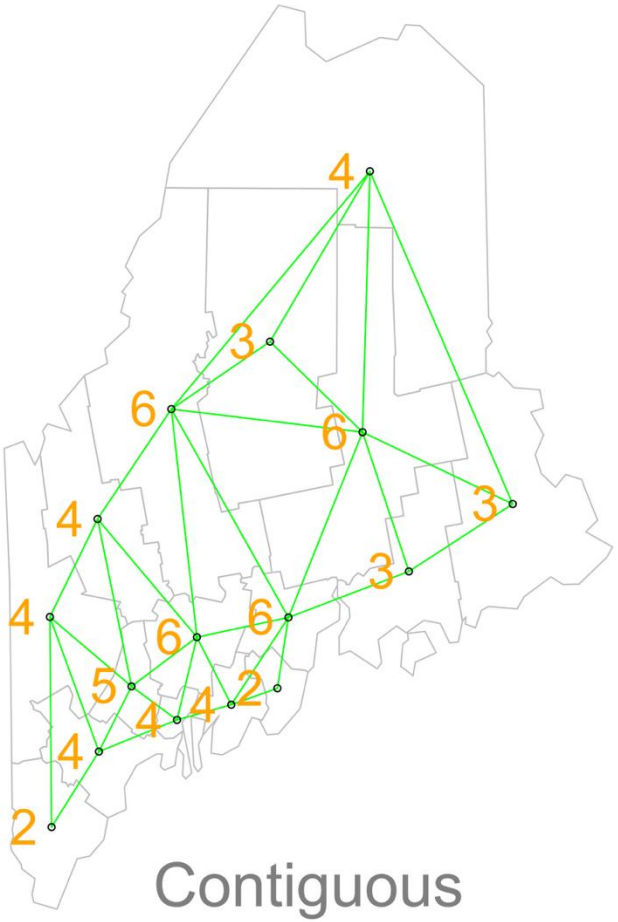
Simulated example: house prices in 100 neighbourhoods. Significant at p of 0.05 or less (199 shuffles per neighbourhood).

Spatial weights matrix

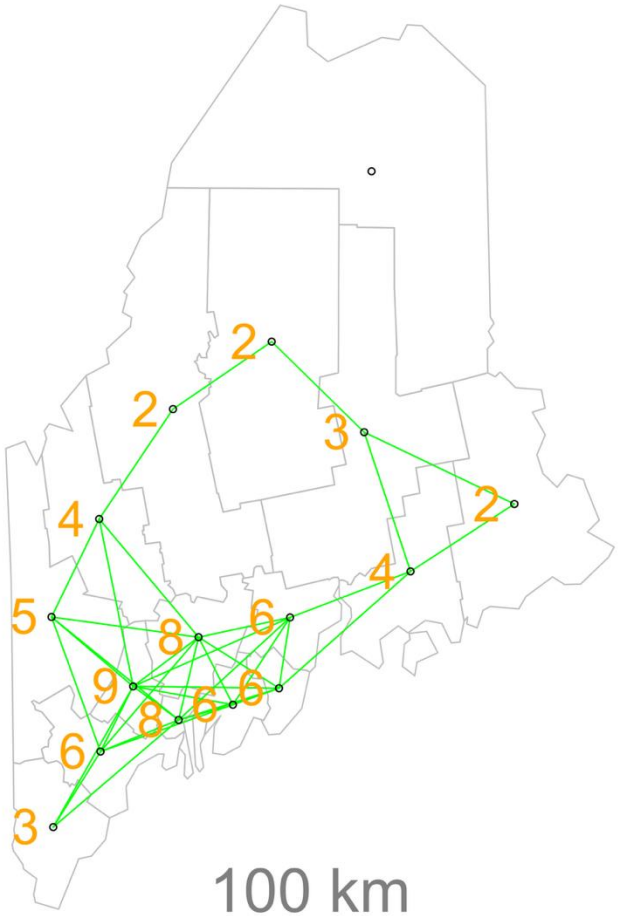
Who is your neighbour?

- To calculate any measure of spatial autocorrelation, we first need to define who counts as a neighbour.
- There are three common ways to define neighbours:
  - Contiguity: neighbours share a boundary, either along edges (rook contiguity) or also at corners (queen contiguity).
  - Fixed distance: neighbours are all the units within a certain distance of a unit.
  - Nearest neighbours: the k closest units, for example the five closest.

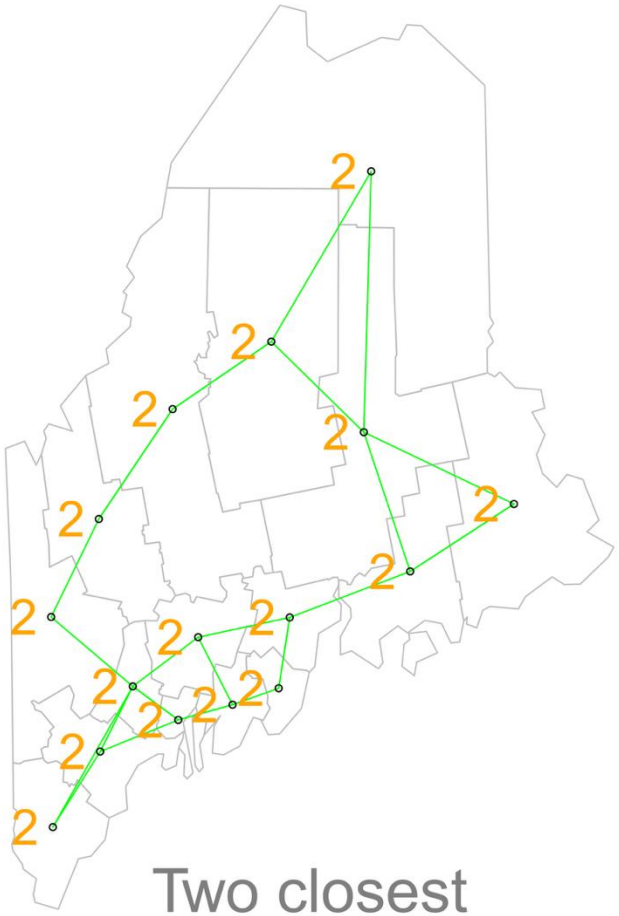
Who is your neighbour?



Contiguous



100 km

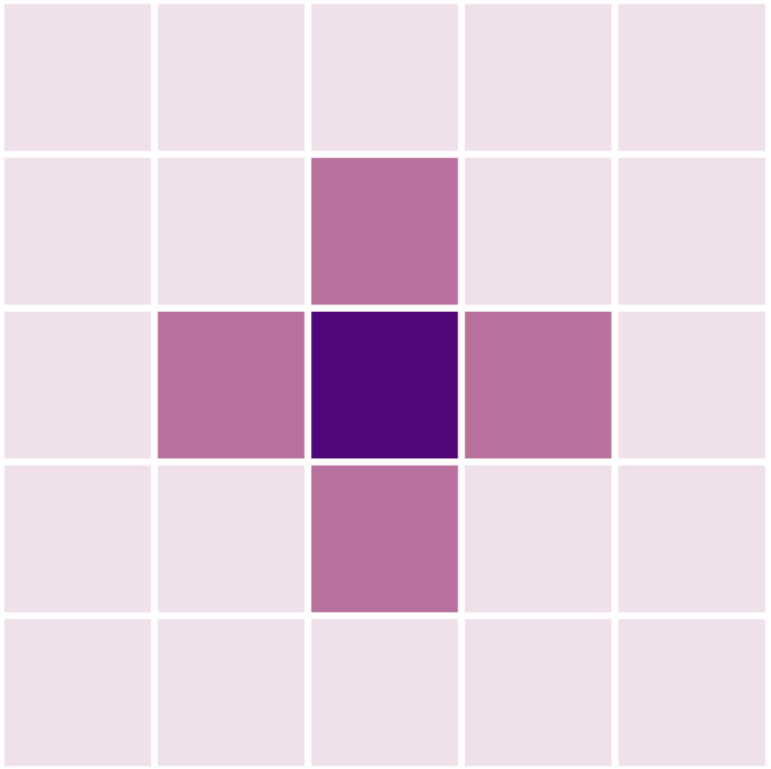


Two closest

Spatial weights matrix

- How to use the relationships between neighbouring features?
- Spatial weights matrix: an `N x N` non-negative matrix (`W`) that summarises all spatial relationships between the features in a dataset.
- It provides a formal expression of spatial dependency by quantifying the influence of each feature on every other feature based on their spatial proximity or other relevant criteria.

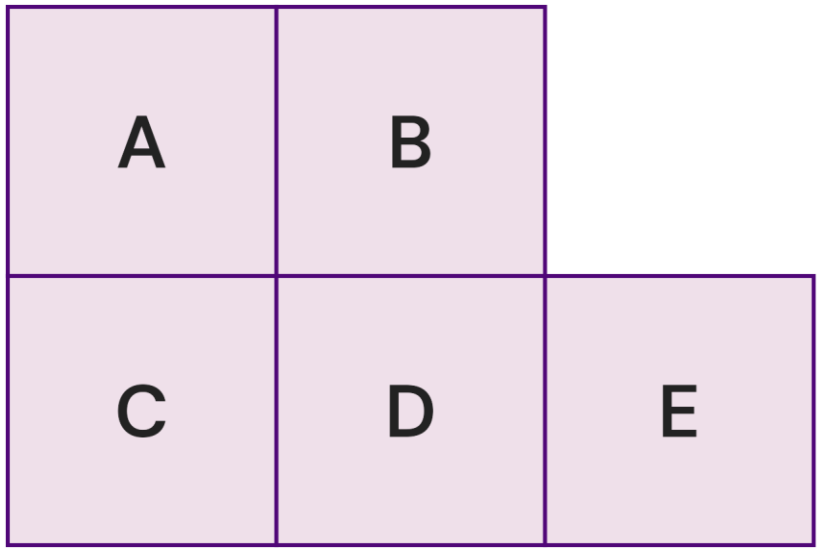
Rook contiguity



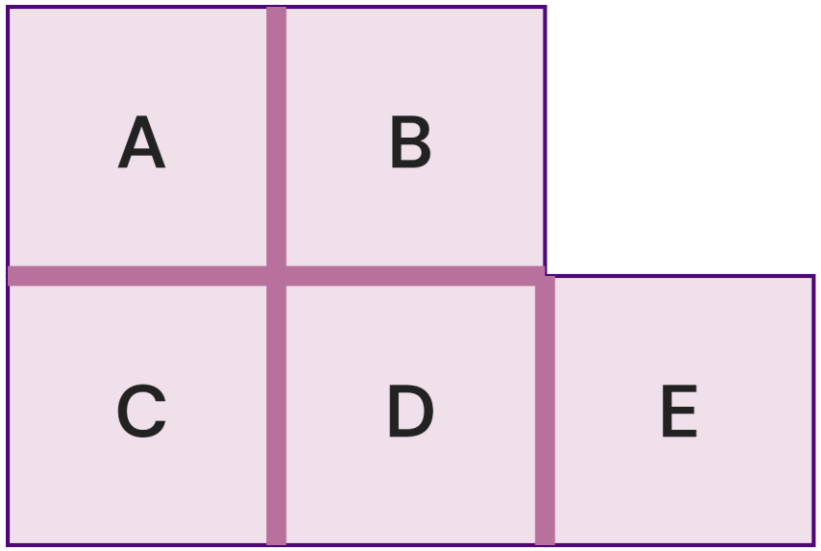
Area of interest

Rook neighbours: share an edge

Rook contiguity



Rook contiguity



Neighbours of each area

A: B, C

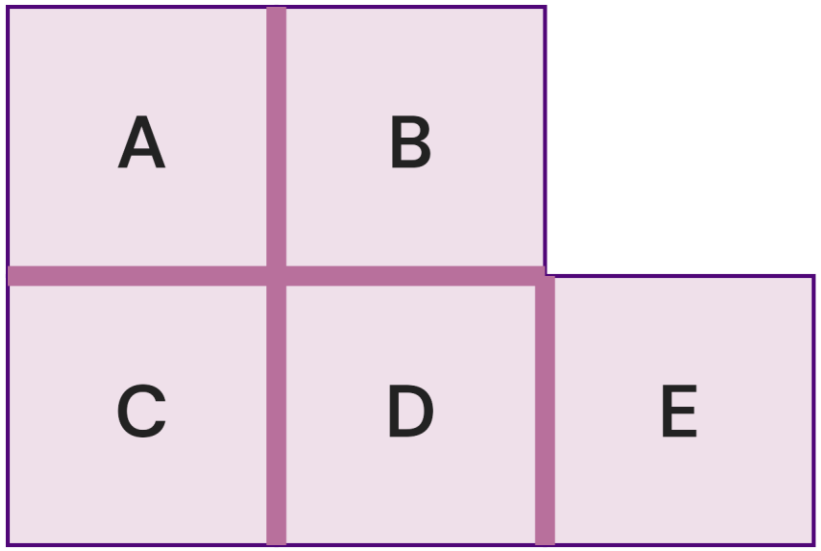
B: A, D

C: A, D

D: B, C, E

E: D

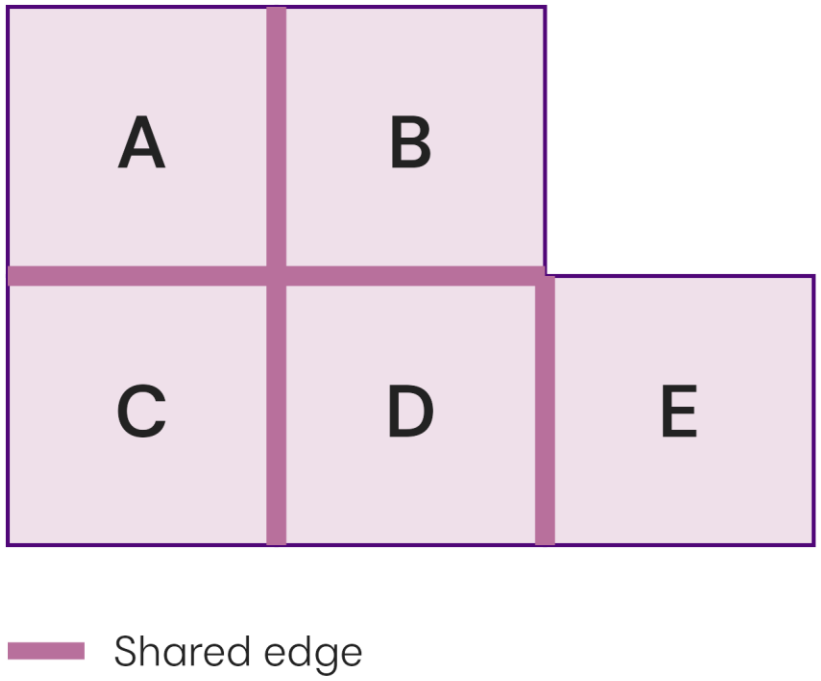
Rook contiguity



Binary weights: 1 = neighbour, 0 = not

	A	B	C	D	E
A	0	1	1	0	0
B	1	0	0	1	0
C	1	0	0	1	0
D	0	1	1	0	1
E	0	0	0	1	0

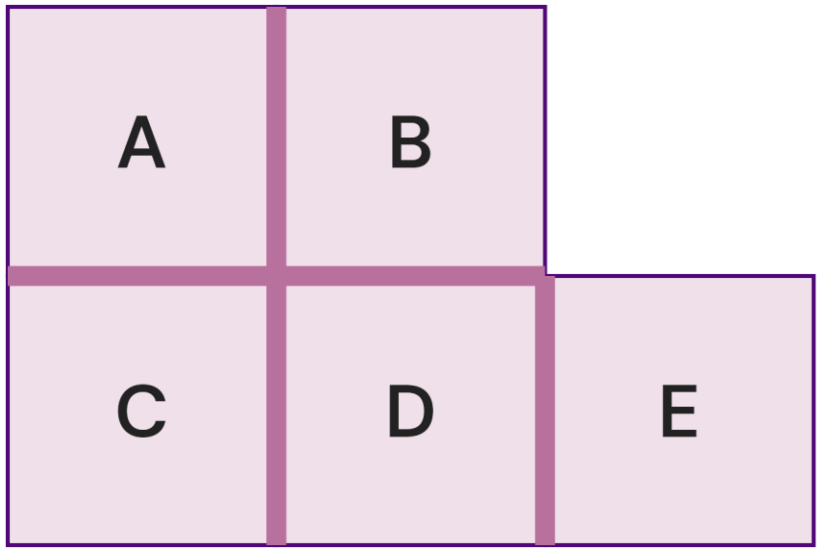
Rook contiguity



w = row sum = number of neighbours

	A	B	C	D	E		w
A	0	1	1	0	0		2
B	1	0	0	1	0		2
C	1	0	0	1	0	→	2
D	0	1	1	0	1		3
E	0	0	0	1	0		1

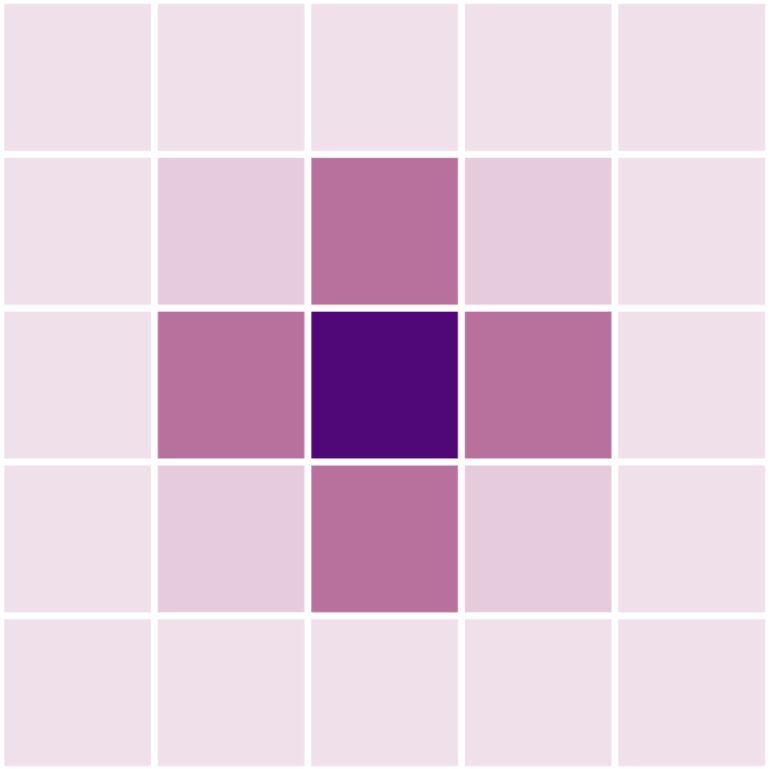
Rook contiguity



Row-standardised: each 1 divided by w

	A	B	C	D	E		w
A	0	.5	.5	0	0		2
B	.5	0	0	.5	0		2
C	.5	0	0	.5	0	→	2
D	0	.33	.33	0	.33		3
E	0	0	0	1	0		1

Queen contiguity

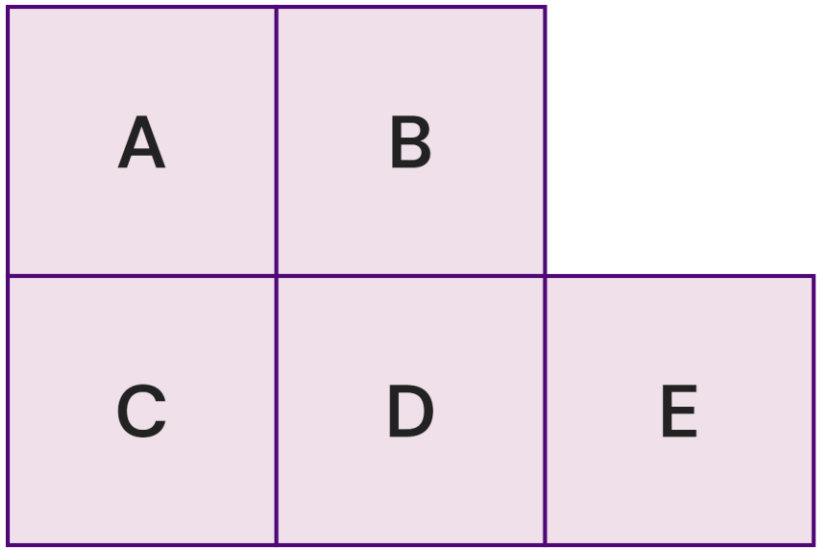


Area of interest

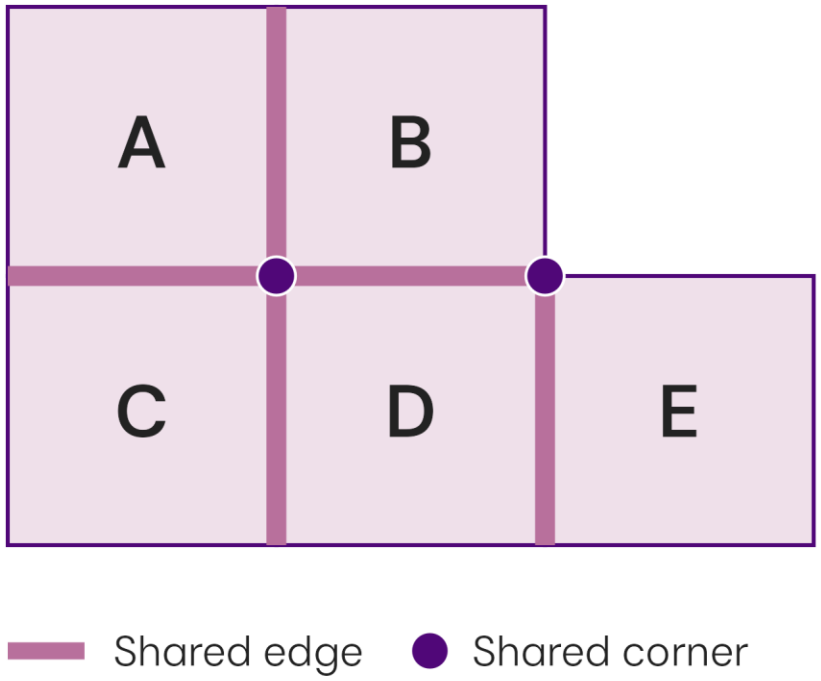
Share an edge (rook)

Also share a corner (queen)

Queen contiguity



Queen contiguity



Neighbours of each area

A: B, C, D

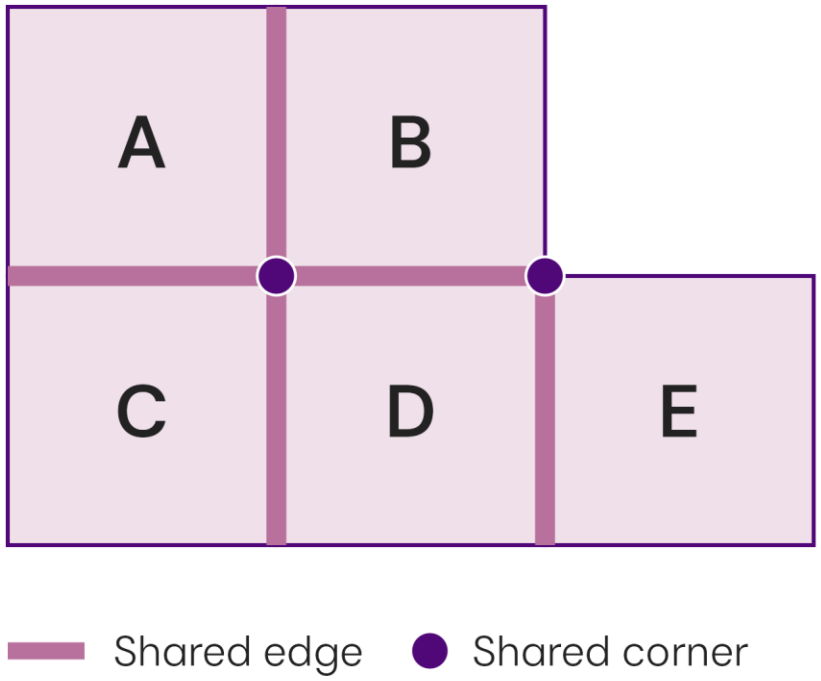
B: A, C, D, E

C: A, B, D

D: A, B, C, E

E: B, D

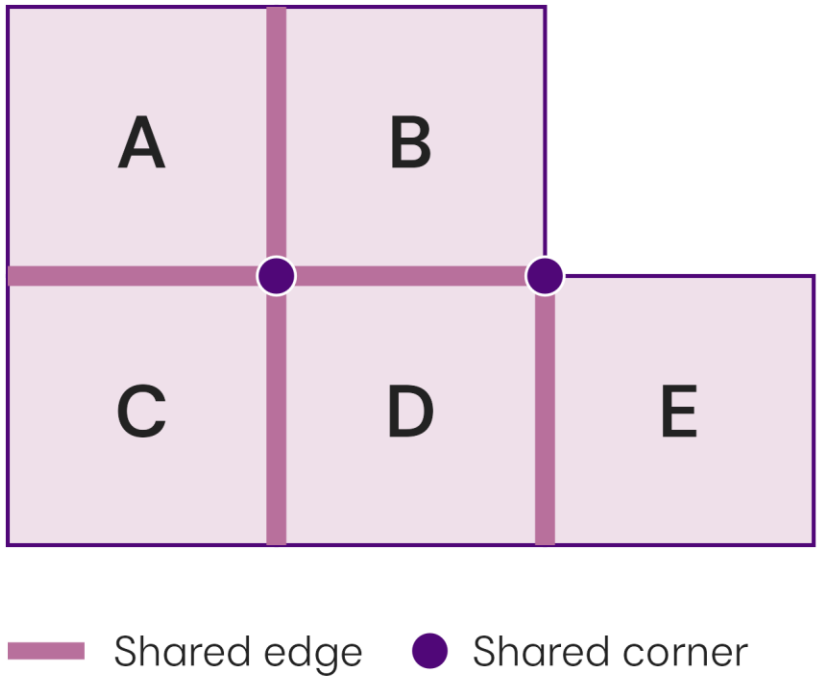
Queen contiguity



Binary weights: 1 = neighbour, 0 = not

	A	B	C	D	E
A	0	1	1	1	0
B	1	0	1	1	1
C	1	1	0	1	0
D	1	1	1	0	1
E	0	1	0	1	0

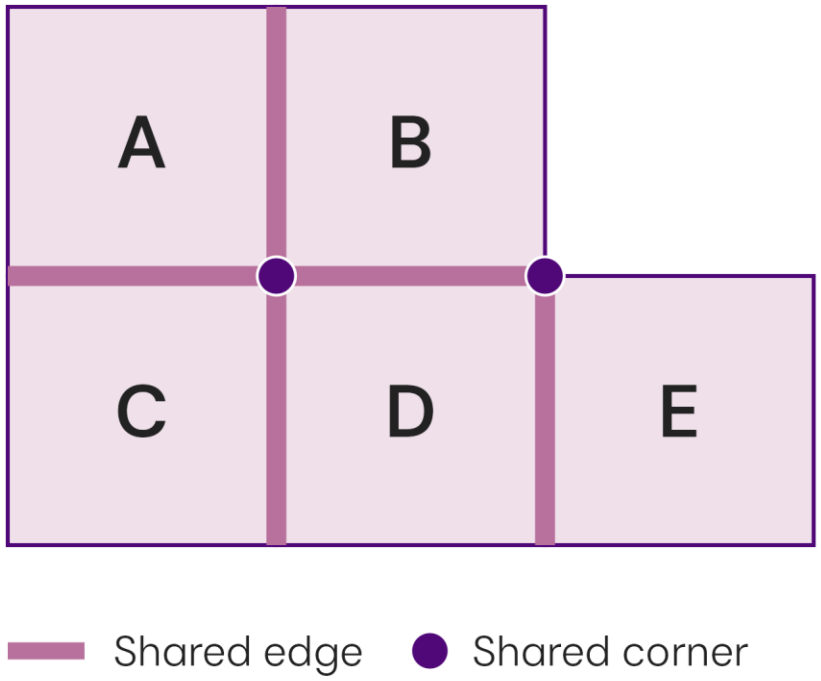
Queen contiguity



w = row sum = number of neighbours

	A	B	C	D	E		w
A	0	1	1	1	0		3
B	1	0	1	1	1		4
C	1	1	0	1	0	→	3
D	1	1	1	0	1		4
E	0	1	0	1	0		2

Queen contiguity

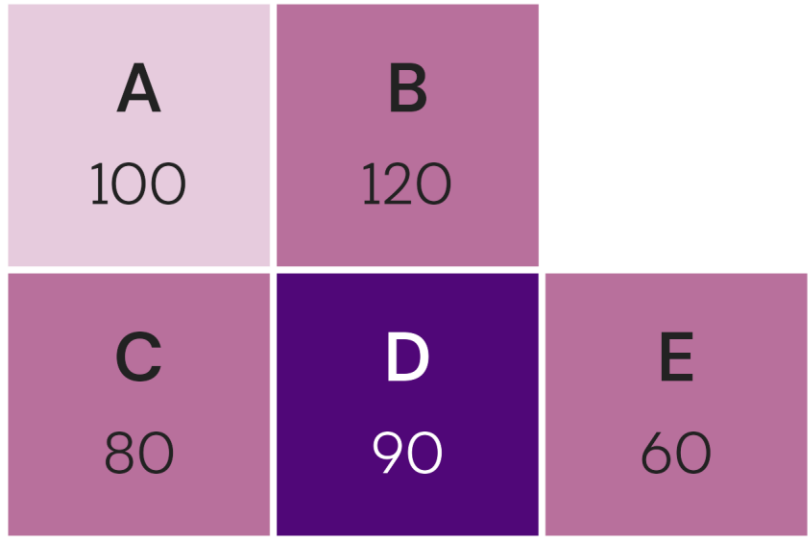


Row-standardised: each 1 divided by w

	A	B	C	D	E		w
A	0	.33	.33	.33	0		3
B	.25	0	.25	.25	.25		4
C	.33	.33	0	.33	0	→	3
D	.25	.25	.25	0	.25		4
E	0	.5	0	.5	0		2

Spatial lag: the average of the neighbours

House price (£000) in each area



Area of interest
Rook neighbours
Added by queen

Lag of D: the average of its neighbours

Rook neighbours: B, C, E

(120 + 80 + 60) / 3 = 86.7

Queen neighbours: A, B, C, E

(100 + 120 + 80 + 60) / 4 = 90

Different neighbours, different lag.

The lag is the vertical axis
of the Moran scatterplot.

Spatial weights matrix

Standardisation can be done in different ways. In `spdep` package:

- “B” coding scheme: no standardisation, every link has weight 1
- “W” coding scheme: row standardisation, each row sums to 1, so the lag is the average of the neighbours
- “C” coding scheme: global standardisation, all weights sum to the number of units
- “U” coding scheme: all weights sum to 1

Topology



Conclusion

- Moran's I gives one value for the whole map, local Moran's I one value for each area.
- Significance: compare the observed value with the values from shuffled data (Monte Carlo).
- Local results show clusters and outliers; with many tests, some will be significant by chance.
- The results depend on how neighbours are defined: contiguity, fixed distance or nearest neighbours.
- Next week: is there spatial autocorrelation left in the residuals of a model?

Questions

Justin van Dijk

j.t.vandijk@ucl.ac.uk

